Grade: Check Plus

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Assignment 1

x	$f(x)$	$xf(x)$	$(x-50)^2 f(x)$
10	0.10	1	160
20	0.15	3	135
40	0.30	12	30
60	0.15	9	15
80	0.25	20	225
100	0.05	5	125
	1.00	50	690

a. What is the probability that a randomly selected customer withdraws at least \$60 from the ATM?

$P(X \geq 60) = P(60) + P(80) + P(100)$

✓ Qa

$0.15 + 0.25 + 0.05 = 0.45$

$\boxed{P = 0.45 \text{ or } 45\%}$

✓ Qb

b. Show by calculation that the expected value withdrawn from the ATM is \$50 and that the standard deviation of the amount withdrawn from this ATM is \$26.27

$\mu = \sum_{\text{all } x} x f(x) = 50$

1. $10 \times .10 = 1$	$40 \times .30 = 12$	$80 \times .25 = 20$
$20 \times .15 = 3$	$60 \times .15 = 9$	$100 \times 0.05 = 5$

$$\mu = 1 + 3 + 12 + 9 + 20 + 5 = \boxed{50}$$

Variance

$$\sigma^2 = \sum_{\text{all } x} (x - 50)^2 P(x) = 690$$

$$(10-50)^2 \times 0.10 = 160$$
$$(20-50)^2 \times 0.15 = 135$$
$$(40-50)^2 \times 0.30 = 30$$
$$(60-50)^2 \times 0.15 = 15$$
$$(80-50)^2 \times 0.25 = 225$$
$$(100-50)^2 \times 0.05 = 125$$

$$\sigma^2 = 160 + 135 + 30 + 15 + 225 + 125 = 690$$

$$\sigma = \sqrt{690} = \boxed{26.27}$$

C. The manager of this convenience store tells you that an expected value of \$50 is not correct because it is impossible to withdraw \$50 from the ATM. How would you answer the manager? ✓ Qc

The expected value does not have to be one of the possible withdrawal amounts. It's describing the long-run average withdrawal across many customers. If you were to monitor a large number of withdrawals, the average amount per transaction would be around \$50.

d. What is the probability that the amount withdrawn from the ATM will fall within one standard deviation of the mean?

$P(50 - 26.27 \leq x \leq 50 + 26.27)$

$P(23.73 \leq x \leq 76.27)$

$P(40 \leq x \leq 60)$

$P(0.30 + 0.15) = \boxed{0.45 \text{ or } 45\%}$

e. What is the probability that the amount withdrawn from the ATM will fall within two standard deviations of the mean?

$P(50 - 2 \times 26.27 \leq x \leq 50 + 2 \times 26.27)$

$P(-2.54 \times 102.54)$

$P(10 \times 100)$

$P(0.10 + 0.15 + 0.30 + 0.15 + 0.25 + 0.05)$

$\boxed{= 1 \text{ or } 100\%}$

✓ Qe

✓ Qd